

KULBHUSHAN CHAND

Senior Project Scientist
Indian Institute of Technology Mandi,
HP, India – 175075

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I am an interdisciplinary researcher working at the intersection of human-computer interaction, extended reality, physiological computing, wearable sensing, and intelligent interactive systems. My research focuses on the design and development of biofeedback technologies, immersive virtual environments, and multimodal interaction techniques for applications in healthcare, disaster resilience, human performance, and assistive technologies. My teaching emphasizes project-based and research-oriented learning supported by interactive tools and open-source technologies.

EDUCATION

Ph.D. <i>Electronics and Communication Engineering, National Institute of Technology Jalandhar</i> <i>Dissertation: Low Cost Multimodal Biofeedback for Mental Relaxation</i> <i>Supervisor: Prof. Arun Khosla</i>	2015 – 2022
M.Tech. <i>Electronics and Communication Engineering, Lovely Professional University</i> <i>Dissertation: A New Approach for ECG Signal Denoising Using Adaptive Filtering</i> <i>University Academic Honor (First in Order of Merit, First Division with Distinction)</i>	2011 – 2013
B.Tech. <i>Electronics and Communication Engineering, Guru Nanak Dev University</i> <i>Graduated with Honors (First Division)</i>	2006 – 2010

ACADEMIC APPOINTMENTS

Senior Project Scientist <i>Centre for Climate Change and Disaster Management (C3DAR), Indian Institute of Technology Mandi</i> <i>Project: Tata Trusts SAFER Hills – Augmented Facilities for Empowering Resilience in Himalayan Hills</i> <i>Advisor: Prof. Varun Dutt</i>	Feb 2026 – Present
Postdoctoral Fellow <i>Technology Innovation Hub, Indian Institute of Technology Mandi</i> <i>Project: Multisensory Human Computer Interaction and Biofeedback</i> <i>Advisor: Prof. Varun Dutt</i>	Nov 2022 – Feb 2026
Predoctoral Fellow <i>Technology Innovation Hub, Indian Institute of Technology Mandi</i> <i>Project: Multisensory Human Computer Interaction and Biofeedback</i> <i>Advisor: Prof. Varun Dutt</i>	Jun 2022 – Nov 2022
Doctoral Researcher <i>Department of Electronics and Communication Engineering, National Institute of Technology Jalandhar</i>	Aug 2015 – Apr 2022
Teaching Assistant (Concurrent) <i>Department of Electronics and Communication Engineering, National Institute of Technology Jalandhar</i>	Aug 2015 – Jul 2020
Project Team Member (Concurrent) <i>ISRO Space Technology Incubation Center (S-TIC), National Institute of Technology Jalandhar</i> <i>Project: Strengthening Ridge to Valley based NRM planning under Mission Water Conservation using Geospatially enabled Serious Gaming</i> <i>Supervisor: Prof. Arun Khosla</i>	Apr 2020 – Mar 2024
Assistant Professor <i>Department of Electronics and Communication Engineering, Lovely Professional University</i>	Jul 2013 – Jul 2015
M.Tech. Researcher <i>Department of Electronics and Communication Engineering, Lovely Professional University</i>	Jul 2011 – Jun 2013
Teaching Assistant (Concurrent) <i>Department of Electronics and Communication Engineering, Lovely Professional University</i>	Jan 2012 – Nov 2012
Lecturer <i>Department of Electronics and Communication Engineering, Lovely Professional University</i>	Dec 2010 – May 2011

Journals (*corresponding author)

- J.6 Kumar, Y., Garg, S., Agarwal, P., **Chand, K.**, Bhavsar, A., & Dutt, V*. (2026). Affordable ECG Sensing: A Machine Learning-Enhanced Framework for Scalable Human–Machine Interaction in Telehealth. *IEEE Sensors Journal*, 26(10), 14539–14546.
- J.5 **Chand, K.***, Chandra, S., & Dutt, V.* (2024). Raga Bhairavi in virtual reality reduces stress-related psychophysiological markers. *Scientific Reports*, 14(1), 24816.
- J.4 **Chand, K.***, Chandra, S., & Dutt, V.* (2024). A comprehensive evaluation of linear and non-linear HRV parameters between paced breathing and stressful mental state. *Heliyon*, 10(11), e32195
- J.3 **Chand, K.***, & Khosla, A. (2022). BioNES: A plug-and-play MATLAB-based tool to use NES games for multimodal biofeedback. *SoftwareX*, 19, 101184.
- J.2 **Chand, K.***, & Khosla, A. (2022). Efficacy of Using Retro Games in Multimodal Biofeedback Systems for Mental Relaxation. *International Journal of Gaming and Computer-Mediated Simulations (IJGCMS)*, 14(1), 1–23.
- J.1 **Chand, K.***, & Khosla, A. (2022). MATLAB-Based Real-Time Data Acquisition Tool for Multimodal Biofeedback and Arduino-Based Instruments: Arduino Firmata Data Acquisition (AfDaq). *Journal of Information Technology Research (JITR)*, 15(1), 1–20.

Conferences - research papers (*corresponding author)

- C.12 Katta, S. S., **Chand, K.***, Aggarwal, P., & Dutt, V.* (2026, October 22). Auditory Environmental Design and Executive Control: Evaluating Background Sound Effects on Interference Resolution in a Human–System Task. ASPIRE: the 69th HFES International Annual Meeting, Nevada, USA. [Accepted]
- C.11 Sonam, S., **Chand, K.**, Mehra, A., Sharma, S., Meena, Y. K., Kaur, G., Rao, A. K., Govindaraji, R., Bhavsar, A., & Dutt, V.* (2026). Sensing and Modulating Autonomic State During a Virtual-Reality Combat Task: A Feasibility Study Toward Closed-Loop XR. NeuroXR 2026, Workshop at the IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2026). [Accepted]
- C.10 Sharma, S., **Chand, K.**, Rao, A. K., & Dutt, V.* (2026). Validation-First NeuroXR: Immersive VR Spatial Navigation for Route-Memory Assessment Against the Groton Maze Learning Test. NeuroXR 2026, Workshop at the IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2026). [Accepted]
- C.9 Agarwal, K., **Chand, K.***, Mehra, A., Uday, K. V., & Dutt, V.* (2026). Flash Flood Exposure and Localized Driving Control in Immersive VR: Evidence from Approach Speed, Braking, and Hazard-Zone Interaction. PRISM 2026. 1st International Conference on Pioneering Research in Intelligent Systems and Morphologies PRISM 2026. [In Press]
- C.8 Mehra, A., **Chand, K.***, Uday, K. V., & Dutt, V.* (2026). Effects of Warning Reliability and Environmental Visibility on Driving Behavior and Physiological Stress in a VR-Based Landslide Hazard Simulation. Proceedings of the 19th International Conference on Pervasive Technologies Related to Assistive Environments. PETRA '26, Crete, Greece. [In Press]
- C.7 Agarwal, K., **Chand, K.***, Mehra, A., Uday, K. V., & Dutt, V.* (2026). Rain-Induced Shifts in Driving Strategy on Flood-Prone Mountain Roads. Proceedings of the 19th International Conference on Pervasive Technologies Related to Assistive Environments. PETRA '26, Crete, Greece. [In Press]
- C.6 **Chand, K.***, Mehra, A., Sonam, S., Tomar, K., Meena, Y. K., Kumar, A., Kaur, G., Rao, A. K., Govindaraji, R., Bhavsar, A., & Dutt, V.* (2026). Asymmetric Behavioral Adaptation from Initial Difficulty Exposure in a VR-Based Shooting Task. Proceedings of the 19th International Conference on Pervasive Technologies Related to Assistive Environments. PETRA '26, Crete, Greece. [In Press]
- C.5 **Chand, K.***, & Dutt, V.* (2026). When Feedback Is Introduced Matters: Short-Term Effects of Biofeedback Sequencing in Stress-Sensitive Gameplay. Proceedings of the 19th International Conference on Pervasive Technologies Related to Assistive Environments. PETRA '26, Crete, Greece. [In Press]
- C.4 **Chand, K.***, Khosla, A., & Dutt, V. (2026). Towards Affordable Biofeedback System for Mental Relaxation. In L. Behera, A. Bandyopadhyay, & G. Ramanath (Eds.), *Cognitive Biomarkers and BCI Applications* (pp. 248–269). Springer Nature Switzerland.
- C.3 Soni, S., **Chand, K.**, & Chowdhury, S. R.* (2025). Development of a Wearable fNIRS System for Non-invasive Cerebral Oxygenation Monitoring. *6th International Conference on Frontiers in Computing and Systems (COMSYS 2025)*, Warsaw University of Technology, Poland. [In Press]
- C.2 Chandra, S., **Chand, K.**, & Dutt, V.* (2023). Impact of Indian Classical Raga in Immersive Environments on Human Psycho-physiological Parameters. *Proceedings of the 16th International Conference on Pervasive Technologies Related to Assistive Environments. PETRA '23*, Corfu, Greece.
- C.1 Rao, A. K., Fatma, Z., Menon, V. K., Bhavsar, A., Chowdhury, S. R., Chandra, S., Dutt, V.*, & **Chand, K.** (2023). Prediction of decision-making performance post-longitudinal tDCS administration via EEG features and machine learning. *Proceedings of the 16th International Conference on Pervasive Technologies Related to Assistive Environments. PETRA '23*, Corfu, Greece.

Conferences – posters & abstracts

- P.3 Katta, S. S., **Chand, K.***, Aggarwal, P., & Dutt, V.* (2026, October 22). Auditory Environmental Design and Executive Control: Evaluating Background Sound Effects on Interference Resolution in a Human–System Task [Poster]. ASPIRE: the 69th HFES International Annual Meeting, Nevada, USA. [Accepted]
- P.3 Sharma, N.*, **Chand, K.**, & Khosla, A. (2026). Technology-Assisted Cognitive Enhancement in ASD: Impact of a Gesture-Based Serious Game Intervention [Poster]. *68th European Autism Congress*, Paris, France. [Accepted]
- P.2 **Chand, K.***, Chandra, S., & Dutt, V. (2025). Raga Bhairavi in virtual reality reduces stress-related psychophysiological markers [Poster]. *Adani Global Indology Conclave*, Ahmedabad, India.
- P.1 **Chand, K.***, Chandra, S., & Dutt, V. (2023). Effects of Relaxed and Stressful Mental States on the Heart Rate Variability Parameters. *Mind, Brain, and Consciousness Conference (MBCC 2023)*, IIT Mandi, India.

Book Chapters

- B.4 Jain, B.*, Khosla, A., **Chand, K.**, & Ahuja, K. (2019). The Pursuit of Zero Food Waste: A Gamified Approach Promoting Avoidance of Dormitory Mess Food Wastage in Educational Institutions. In *Global Initiatives for Waste Reduction and Cutting Food Loss* (pp. 243–267). IGI Global.
- B.3 Mukherjee, A.*, Goyal, P., Singh, A., Khosla, A. K., Ahuja, K., & **Chand, K.** (2019). Pursuit of Research: A Gamified Approach Promoting Research Engagement Among Undergraduate Students. In *Fostering Multiple Levels of Engagement in Higher Education Environments* (pp. 132–155). IGI Global.
- B.2 Padmakar, G. S. V.*, Khosla, A., & **Chand, K.** (2019). LearnEasy-Android Application as a Technological Intervention for Children With Dyslexia. In *Emerging Trends in the Diagnosis and Intervention of Neurodevelopmental Disorders* (pp. 236–248). IGI Global.
- B.1 Setia, D.*, Singh, R., Sharma, A., Khosla, A., Ahuja, K., & **Chand, K.** (2019). Enhancing Tourism and Cultural Experience Through Gamification. In *Positioning and Branding Tourism Destinations for Global Competitiveness* (pp. 152–171). IGI Global.

Manuscripts in Review

- M.1 Mehra, A., **Chand, K.**, Rao, A. K., Gonsalves, T., Meyer, J., Uday, K., & Dutt, V. (2026). A Triad Analysis of Active vs. Passive Warning Cues in VR Landslide Driving [Manuscript in review]. ISMAR 2026.

CONFERENCE PRESENTATIONS (PEER-REVIEWED)

When Feedback Is Introduced Matters: Short-Term Effects of Biofeedback Sequencing in - <i>Stress-Sensitive Gameplay</i> 19 th Pervasive Technologies Related to Assistive Environments (PETRA) 2026 Crete, Greece	July 2026
Raga Bhairavi in virtual reality reduces stress-related psychophysiological markers <i>Adani Global Indology Conclave</i> Ahmedabad	Nov 2025
Towards Affordable Biofeedback System for Mental Relaxation <i>Mind, Brain, and Consciousness Conference (MBCC) 2025</i> Indian Institute of Technology Mandi	Jun 2025
Effects of Relaxed and Stressful Mental States on the Heart Rate Variability Parameters <i>Mind, Brain, and Consciousness Conference (MBCC) 2023</i> Indian Institute of Technology Mandi	Dec 2023

INTELLECTUAL PROPERTY RIGHTS

Patents

HRV based biofeedback device <i>Status: Invention Disclosure Form submitted at Indian Institute of Technology Mandi</i> <i>Description: An adaptive HRV based biofeedback device for mental stress management</i>	Dec 2025
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Copyrights

Biofeedback Nintendo Entertainment System (BioNES) <i>Registration No.: SW-2025021242 Copyright Office, Government of India</i> <i>Description: MATLAB-based framework for integrating NES games into multimodal biofeedback systems for mental wellness research.</i>	Aug 2025
Realtime arduino based HRV biofeedback system for mental stress assessment <i>Registration No.: SW-2025021178 Copyright Office, Government of India</i> <i>Description: Arduino code for a low-cost HRV-based mental stress assessment device with real-time IBI acquisition, artifact correction, HR, RMSSD, SDNN, and stress percentage estimation.</i>	Aug 2025
Arduino Firmata Data Acquisition (AfDaq) <i>Registration No.: SW-2025021174 Copyright Office, Government of India</i> <i>Description: Open-source MATLAB tool for real-time multichannel data acquisition and visualization from Arduino boards, used in biofeedback systems and device control.</i>	Aug 2025

PROJECTS

Curricular Projects and Teaching Tools

Arduino Workshop for Kids E-Book <i>National Institute of Technology Jalandhar</i>	2022
Gamification (ECX-488) Course E-Book, 3 Credits <i>National Institute of Technology Jalandhar</i>	2021
Classroom Gamification, Software Tool, 3 Credits <i>National Institute of Technology Jalandhar</i>	2018
Gamification (ECX-488), Course, 3 Credits <i>National Institute of Technology Jalandhar</i>	2016

Open-Source Tools and Software

AfDaq: A plug and play MATLAB based tool for biofeedback and arduino based instruments https://github.com/kulbhushanchand/AfDaq	2021
BioNES: A plug-and-play MATLAB based tool to use NES games for multimodal biofeedback https://github.com/kulbhushanchand/BioNES	2021
MCP4251: Arduino library for MCP4251 Digital Potentiometer https://github.com/kulbhushanchand/MCP4251	2021
classroom-gamification: Tool to gamify your classroom with an online leaderboard https://github.com/kulbhushanchand/classroom-gamification	2018

INVITED TALKS / EXPERT LECTURES / WORKSHOPS

Invited Talks

Human Motivation and Engagement Analysis using Octalysis Gamification Framework <i>5th Winter School on Cognitive Modeling - 2023</i> <i>Indian Institute of Technology Mandi</i>	Oct 2023
Human Motivation and Engagement Analysis using Octalysis Gamification Framework <i>Winter School on Cognitive Modeling - 2022</i> <i>Indian Institute of Technology Mandi</i>	Oct 2022
Arduino and Open-Source Technologies <i>DAV Institute of Engineering and Technology</i>	Jun 2019
Octalysis <i>AICTE ATAL Sponsored Online Faculty Development Program on Gamification</i> <i>National Institute of Technology Jalandhar</i>	Nov 2020

Expert Lectures

From Sensors to Systems: Building HRV-Based Biofeedback for Mental Health <i>Guest Lecture for undergraduate students in School of Engineering Design and Automation-Engineering</i> <i>GNA University</i>	Apr 2026
The Rhythm of the Heart: HRV, Autonomic Regulation, and Applications <i>Guest Lecture for open elective course Music and Cognition (IK 515) in IKSMHA Center</i> <i>Indian Institute of Technology Mandi</i>	Apr 2026
Octalysis Gamification Framework <i>Guest Lecture for undergraduate course Gamification (ECOE-496)</i> <i>National Institute of Technology Jalandhar</i>	Oct 2025
Octalysis Gamification Framework <i>Guest Lecture for undergraduate course Gamification (ECOE-496)</i> <i>National Institute of Technology Jalandhar</i>	Feb 2023
Octalysis Gamification Framework <i>Guest Lecture for undergraduate course Gamification (ECX-488)</i> <i>National Institute of Technology Jalandhar</i>	Sep 2022

Workshops

Arduino workshop for kids <i>Jalandhar</i>	Nov 2017
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RESEARCH FUNDING

On-Device Multimodal Sensing for Real-Time Training, Attention Regulation, and Performance Adaptation <i>Qualcomm XR University Research Funding Program</i>	2026
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<i>Status: Under Review (Co-PI)</i>	
Motion-Seat XR for Landslide Decision-Making, Active Warnings, and On-Device Perception <i>Qualcomm XR University Research Funding Program</i> <i>Status: Under Review (Co-PI)</i>	2026
Immersive VR and Multimodal Sensing for EV Charging Cybersecurity Human Factors <i>Qualcomm XR University Research Funding Program</i> <i>Status: Under Review (Co-PI)</i>	2026

HONORS AND AWARDS

University Academic Honor (M.Tech. – First in Order of Merit, First Division with Distinction) <i>Lovely Professional University</i>	2013
Graduated with Honors (B.Tech. – First Division) <i>Guru Nanak Dev University</i>	2010

COMPETITIVE QUALIFICATIONS

UGC-NET Qualified with JRF <i>University Grants Commission</i>	2013
GATE-2013 Qualified <i>Ministry of Education</i>	2013
GATE-2011 Qualified <i>Ministry of Education</i>	2011

ACADEMIC SERVICE

Mind, Brain, and Consciousness Conference (MBCC 2023) <i>Indian Institute of Technology Mandi, Role: Workshop Organizer</i>	Dec 2023
Hackathon 2022 <i>Indian Institute of Technology Mandi, Role: Jury member</i>	Jul 2022
Toycathon 2021 <i>Ministry of Education, Role: Primary Evaluator</i>	Jan 2021
ATAL Online FDP on Gamification <i>National Institute of Technology Jalandhar, Role: Assisted in conduction</i>	Nov 2020
GIAN Course on Autism and Assisted Technologies <i>National Institute of Technology Jalandhar, Role: Assisted in conduction</i>	Dec 2018
Sports Meet – 2008 <i>Guru Nanak Dev University, Role: Event Coordinator (Volleyball)</i>	Sep 2008

MEDIA COVERAGE

Raga Bhairavi can lower stress levels: Study explores impact of Indian classical music <i>The Telegraph India, Print and online newspaper</i>	Nov 2024
Young minds dwell on e-concepts <i>The Tribune India, Print and online newspaper</i>	Nov 2017

PROFESSIONAL DEVELOPMENT

Certifications

The Recreational UAS Safety Test (TRUST) <i>Academy of Model Aeronautics</i>	Nov 2022
The Octalysis Framework - Level 1 <i>The Octalysis Group</i>	Dec 2017

Conference (Attended)

Mind, Brain, and Consciousness Conference (MBCC 2025) <i>Indian Institute of Technology Mandi</i>	Jun 2025
CHI 2025 <i>Yokohama, Japan</i>	Apr 2025
Mind, Brain, and Consciousness Conference (MBCC 2023) <i>Indian Institute of Technology Mandi</i>	Dec 2023

MOOC Courses (with certificates)

IITM – Finland – Certificate course on Mastering VR : Fundamentals to Practice <i>Indian Institute of Technology Madras</i>	Oct 2024
DRONE01x: Drones for Agriculture: Prepare and Design Your Drone (UAV) Mission <i>Wageningen University & Research</i>	Sep 2022

Statistics in Medicine <i>Stanford University</i>	Mar 2020
Computer Science 101 <i>Stanford University</i>	Mar 2020
Introduction to CAD: Learn Fusion 360 in 90 minutes <i>Autodesk</i>	Sep 2019
Vital Signs: Understanding What the Body Is Telling Us <i>University of Pennsylvania</i>	May 2018
Advanced MATLAB for Scientific Computing <i>Stanford University</i>	Dec 2017
Introduction to Game Development <i>Michigan State University</i>	Mar 2017
Calculus One <i>The Ohio State University</i>	May 2013
Image and video processing <i>Duke University</i>	Mar 2013
Short Term Courses/Workshops/Webinars (Attended)	
Evaluating Impact Through Simulation & Games <i>Shri Vaishnav Vidyapeeth Vishwavidyalaya</i>	Aug 2025
Learning from Simulations and Games: Guidance from Theory, Concept and Practice <i>Shri Vaishnav Vidyapeeth Vishwavidyalaya</i>	Nov 2024
The Impact of AI on Gamification and Serious Games <i>Shri Vaishnav Vidyapeeth Vishwavidyalaya</i>	Sep 2024
Development of Simulation Game, “Project PAL” using the SECI Model and Collaborative Learning Methods <i>Shri Vaishnav Vidyapeeth Vishwavidyalaya</i>	Jul 2024
Hands on Training and Showcasing: High Density EEG Devices <i>Indian Institute of Technology Mandi</i>	Mar 2024
Simulation Games: A Bridge between Knowledge and Behaviour <i>Shri Vaishnav Vidyapeeth Vishwavidyalaya</i>	Jan 2024
Machine-Learning advances in Human-Computer Interaction <i>Indian Institute of Technology Mandi</i>	Jul 2022
Innovation, IPR and Start-ups <i>National Institute of Technology Jalandhar</i>	Jul 2022
LaTeX for Technical Writing <i>Indian Institute of Technology Mandi</i>	June 2022
The Education Factory <i>National Institute of Technology Jalandhar</i>	Sep 2020
Machine Learning Using Python <i>National Institute of Technology Jalandhar</i>	Sep 2020
Computational Cardiac Engineering - Predictive Modelling and Innovative Sensing Systems <i>Rajasthan Technical University</i>	Sep 2020
IP Commercialization & Technology Transfer <i>National Institute of Technology Jalandhar</i>	Feb 2020
Recent Trends in Biomedical Signal Processing and VLSI Design <i>National Institute of Technology Jalandhar</i>	May 2019
Data Science <i>National Institute of Technology Jalandhar</i>	May 2019
Current Avenues of Research in Electronics and Communication Engineering <i>National Institute of Technology Jalandhar</i>	Mar 2019
Autism and Assisted Technologies (Gian Course) <i>National Institute of Technology Jalandhar</i>	Dec 2018
Brain Science & Technology: Tools and Techniques to understand the Human Brain <i>Indian Institute of Technology Delhi</i>	Dec 2016
How to be your Child’s Occupational therapist <i>SOCH Educational Society</i>	Sep 2016
Biology for Engineers <i>Indian Institute of Technology Bombay</i>	Jun 2016

PROFESSIONAL MEMBERSHIP

Association for Computing Machinery (ACM)
Professional Member

Aug 2023 – Jul 2024; Feb 2026 – Jan 2031

Institute for Systems and Technologies of Information, Control and Communication (INSTICC) Jan 2026 – Dec 2026
Regular Member

EDITORIAL AND PEER REVIEW ROLES

Editorial Review Board Member

IGI Global: International Journal of Gaming and Computer-Mediated Simulations (IJGCMS); International Journal of Game-Based Learning (IJGBL); International Journal of Distance Education Technologies (IJDET)

Ad-Hoc Reviewer

Springer Nature: Discover Psychology; Signal, Image and Video Processing

Cell Press: iScience; Heliyon

Elsevier: Computers in Human Behavior Reports; SoftwareX; HardwareX; Biomedical Signal Processing and Control

Sage: Games for Health Journal

IGI Global: International Journal of Gaming and Computer-Mediated Simulations (IJGCMS); International Journal of Game-Based Learning (IJGBL); International Journal of Distance Education Technologies (IJDET)

ACM: BIOSIGNALS 2026, CHI 2025, CHI 2024

INDUSTRY EXPERIENCE & ENTREPRENEURSHIP

T4A Labs Pvt Ltd., Incubated in NITJ Technology Business Incubator

Apr 2022 – Present

Role: Co-Founder, Works on technologies for autistic kids and autism.

JCT Limited, Filament Unit, Hoshiarpur (Punjab)

Jan 2010 – Jun 2010

Role: Industrial Trainee

JCT Limited, Filament Unit, Hoshiarpur (Punjab)

Jun 2008 – Jul 2008

Role: Industrial Trainee

SKILLS

Programming: Assembly, C, C++, R, Lua, MATLAB, Python, Visual Basic, Pure Data

Applications: NI Multisim, KiCad, Unity3D, RStudio, JASP, Jamovi, Autodesk Fusion 360

Operating Systems: Windows, Ubuntu

Others: Octalysis Framework (Level 1 Certified), Physiological Signal Processing Analysis

SUBJECTS TAUGHT

Teaching Assistant (PhD Student)

2015 – 2020

National Institute of Technology Jalandhar

Gamification, ECX-488

Spring 2020

Scientific Computing Lab, ECPC-351

Spring 2020

Gamification, ECX-488

Fall 2019

Scientific Computing Lab, ECPC-351

Fall 2019

Gamification, ECX-488

Spring 2019

Electronics for Chemists Lab, CY-592

Spring 2019

Gamification, ECX-488

Fall 2018

Analysis and Synthesis of Networks Lab, ECX-211

Fall 2018

Gamification, ECX-488

Spring 2018

Electronics for Chemists Lab, EC-636

Spring 2018

Gamification, ECX-488

Fall 2018

Analysis and Synthesis of Networks, ECX-201

Fall 2018

Analysis and Synthesis of Networks Lab, ECX-211

Fall 2018

Gamification, ECX-488

Spring 2017

Basic Electronics Lab, ECX-102

Spring 2017

Gamification, ECX-488

Fall 2016

Analysis and Synthesis of Networks Lab, ECX-211

Fall 2016

Gamification, ECX-488

Spring 2016

Basic Electronics Lab, ECX-102	Spring 2016
Basic Electronics Lab, ECX-102	Fall 2015
Assistant Professor <i>Lovely Professional University</i>	2013 – 2015
Digital Signal Processing, ECE300	Spring 2015
Digital Signal Processing Laboratory, ECE324	Spring 2015
Analytical Skills-II, PEA302	Spring 2015
Adaptive Signal Processing, ECE531	Fall 2014
Signal and Systems, ECE220	Fall 2014
Basic Simulation Laboratory, CSE222	Fall 2014
Electronic Devices and Circuits, ECE206	Summer 2014
Digital Signal Processing, ECE300	Spring 2014
Probability and Stochastic Processes, ECE256	Spring 2014
Digital Signal Processing Laboratory, ECE324	Spring 2014
Electronic Devices and Circuits, ECE206	Fall 2013
Basic Electrical Engineering, ECE133	Fall 2013
Analog and Linear Integrated Circuits Laboratory, ECE317	Fall 2013
Basic Electrical Engineering Laboratory, ECE134	Fall 2013
Teaching Assistant (MTech student) <i>Lovely Professional University</i>	2011 – 2013
Introduction to Practical Electronics, ECE100	Fall 2012
Electrical and Electronics Workshop, ECE130	Spring 2012
Lecturer <i>Lovely Professional University</i>	2010 – 2011
Microcontroller and Applications, ECE310	Spring 2010
Introduction to Practical Electronics, ECE100	Spring 2010
Unified Electronics Laboratory – II, ECE208	Spring 2010

SPORTS ACTIVITIES

Annual Faculty Sports Meet, 100 mt. (Gold), 200 mt. (Gold) <i>Lovely Professional University</i>	2015
Annual Faculty Sports Meet, 200 mt. (Bronze), 400 mt. (Silver) <i>Lovely Professional University</i>	2014
Annual Faculty Sports Meet, 100 mt. (Gold) <i>Lovely Professional University</i>	2011

LANGUAGES

English – Proficient (Speak, Read, Write, Listen)
Hindi – Native-equivalent (Speak, Read, Write, Listen)
Punjabi – Native (Speak, Read, Write, Listen)

REFERENCES

Prof. Varun Dutt

Professor, School of Computing and Electrical Engineering, Indian Institute of Technology Mandi, Mandi, Himachal Pradesh, India – 175075
varun@iitmandi.ac.in
Professional Engagement: Post Doctoral Advisor

Prof. Arun Khosla

Professor, Department of ECE, Dr B R Ambedkar National Institute of Technology, Jalandhar, India – 144011
khoslaak@nitj.ac.in
Professional Engagement: PhD Supervisor

Prof. Mamta Khosla

Professor, Department of ECE, Dr B R Ambedkar National Institute of Technology, Jalandhar, India – 144011
khoslam@nitj.ac.in
Professional Engagement: Former Head of Department